DALAS Project
Drug Repurposing

Vladislav Antipin (21242244)
Paul Béglin (21408798)
Master of Computer Science, Sorbonne University

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Contents

1 Introduction

1.1 Problem Statement

Clearly define the problem your project addresses. Explain its relevance in the scientific or practical context.

The discovery of new drugs is, although crucial, an extremely expensive and time-consuming process. Many candidate drugs ultimately turn out to be not only ineffective, but also potentially toxic. It is thus of interest to repurpose already approved drugs for diseases they have not yet been tested against. To prioritize potential clinical trials, in addition to experts' knowledge, it's important to be able to predict the likelihood of success for such studies. This project aims to explore the data and develop a potential prediction pipeline for this purpose. We acknowledge that industrial and academic drug development employ far more comprehensive methods, this project, by contrast, focuses on a preliminary, data-driven approach and is exploratory in nature.

1.2 Motivation and Objectives

Explain why this problem is important. State your main objectives and hypotheses.

The ability to predict the success of drug repurposing has academic and clinical value, it helps reducing the experimental cost and accelerating the discovery of effective treatments. The main objectives of this project are to:

- Explore public datasets to identify some relevant features
- Develop a preliminary machine learning pipeline for success prediction
- Evaluate the model performance
- Identify biases and potential ways for improvement

We decided to specify a group of diverse but closely related diseases - immune system disorders and drugs that are approved for at least one member of this group. Figure 1 summarizes pipeline that will be presented in detail in the following sections.

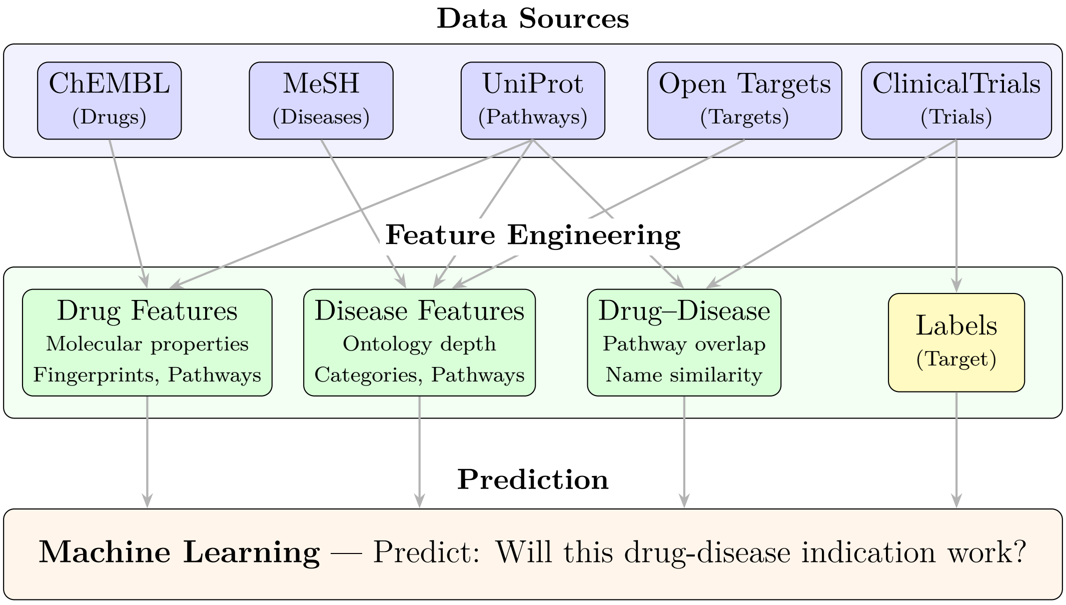

Figure 1: Complete pipeline overview for drug repurposing prediction.

1.3 Biological Rationale: The Drug–Pathway–Disease Network

One of th key assumptions underlying drug repurposing is that drugs and diseases are connected through shared biological pathways. Figure 2 illustrates this relationship as a tripartite graph with three layers.

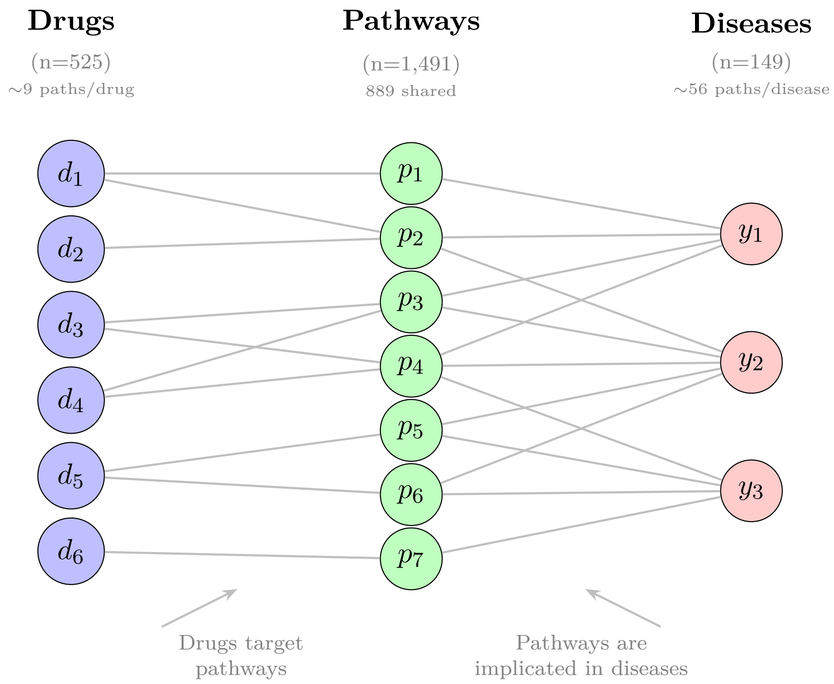

Figure 2: Tripartite graph representing the biological relationship between drugs, pathways, and diseases. Edges connecting drugs and diseases indicating clinical success are not represented.

This tripartite structure motivates our `path_similarity` feature: we compute the

Jaccard similarity between the set of pathways associated with a drug and those associated with a disease. High similarity suggests the drug affects biological processes relevant to the disease, making it a stronger repurposing candidate. So pathway-mediated connections provide mechanistic justification for predictions. However it is important to highlight that this approach is extremely simplified and doesn't capture the directionality of interactions (e.g. positive vs. negative regulation) as well as complex interactions and feedback loops inside and between those pathways.

2 Data

2.1 Data Sources

Describe where the data comes from, collection methods, and aggregation process.

Among the vast amount of medical, pharmacological and chemical publicly available data, we focused our attention on some being able to provide us information on basic classification of diseases (MeSH), drug chemical descriptors and targets (ChEMBL), disease targets (Open Targets) as well as information on drug-disease indication studies (ClinicalTrials). Table 1 summarizes information on data sources. Although initially we were using the web scraping to get information on disease classification, we found out that this way of getting medical information is rather deprecated, so we queried the corresponding databases instead. Data concerning drugs and diseases was agglomerated using the cross references furnished by the same databases (ChEMBL and Open Targets). We used as well a UniProt identifier mapper that allowed us to unify synonymous target IDs as well as enrich them with corresponding molecular pathway identifiers from Reactome database.

Table 1: Description of data sources.

Source	Access Method	Data Retrieved
MeSH RDF	SPARQL endpoint	Disease IDs and ontology
ChEMBL	REST API (Python client with query-style filters)	Drug chemical properties, targets and indications
UniProt	REST API (Python client)	Map target and pathway IDs
Open Targets	GraphQL API	Disease targets
ClinicalTrials.gov	REST API	Trial results and metadata

2.2 Target variable definition

Start with explaining how you got the labels.

The prediction target is a binary success / fail variable indicating whether there was shown an effect or the absence thereof of a drug against a disease. So we needed to obtain binary success labels for each clinically tried drug-disease combination. Drugs that have passed the fourth (last) clinical trial phase for a given disease are automatically labelled as successful. However, other cases are not as obvious. If the last phase wasn't (yet) passed, we can understand the reason by addressing the corresponding clinical trials. There are several possibilities: either this combination was never tried (which is most often the case); or it was tried and shown to be not-effective or effective but for earlier phases; or finally simply because there are no results published or the study was terminated for various (often financial) reasons. There exists some manually and automatically annotated clinical trial databases [Gao et al., 2024] [Chen et al., 2025], but the problem is that either they are pretty limited in time span (e.g. last 10 years) or their "non-success" label encompasses simultaneously trials failed due to drug inefficiency and to other reasons. So we decided to automatically label drug-disease indications that have been tried at least once using following rule:

- If phase 4 is passed, it's definitely a success
- If there is clinical trial evidence, check out results
  - access success of the trial based on results p-values
  - trial is successful if median p-value is below 0.05, fail if above 0.2, not known otherwise
  - if the hightest tried phase is at least 3 and number of successes there is twice as big as fails, assign success to the indication
  - otherwise if the latests phases has twice more fails than successes, assign fail
- If still not decided, check whether this indication was abandoned - the latest date available for any clinical trial concerning this indication was before 2020. We label such indication as fail.

It is interesting to note that most of undecided indications and clinical trials without published results happen to be approximately before 2000 and after 2010, as shown in Figures A1 and A2. One can speculate that it is linked to the fact that ClinicalTrials.gouv was launched in 2000 and it became mandatory to publish there all studies (but not full results) since 2007. As can be seen in Figure **??**, among indictions with no first trial start date available, most are labeled successful which actually correpsonds to indications established long before the launching of ClinicalTrials.gouv.

It's important to note that some of those choices are arbitrary and this probably constitutes the most important weakness of this project.

2.3 Feature Engineering and Description

Provide a table or list explaining each feature and its meaning. Optionally, include relationships between features.

Detailed description of extracted features can be found in Table A1. Categorical features were all one-hot encoded. Most drug features describe chemical and structural properties of those drugs, a well as some andimistration information. Among engendered features are Morgan molecular fingerprints with 2 atoms radius obtained with RDKit module from drug's SMILES (flat string representation of the structure) and yielding a 2048 dimensional bit vector it dimension coding for presence or absence of a particular substructure. To reduce the dimensionality, first 100 dimensions were taken and called `FP_#`.

Disease features are comprised of disease ontology description, one of them being ontological category in addition to immune system diseases, called `cat_#`, for example some of immune diseases are as well neoplastic (e.g. leukemia).

For both dieses and drugs, we obtained a list of molecular (we chose protein) targets - molecules upon which a drug acts or that are involved in the pathogenesis of a diseases. Since targets themselves are extremely heterogenous and very intertwined into molecular pathways, we converted a list of targets for each drug or disease into a list of corresponding pathways. We further performed TF-IDF vectorizations on pathway lists separately for drugs and for diseases. TF-IDF helps downweighting housekeeping pathways (very common ones) and highlights informative ones. We then performed a PCA and extracted first 50 principal components for drugs and 50 for diseases, called as `drug_path_#` and `disease_path_#`. This information captures drug-drug and disease-disease mechanistic similarity based on shared pathways.

To capture the drug-disease pathway similarity, we computed a Jaccard similarity index based on corresponding pathway ID lists. We wanted to include as well drug-disease name similarity based on PubMedBERT embeddings, which reflects how related a drug and a disease in a scientific literature, but later we realized that including this feature for model training would be a mistake since PubMedBERT is trained on recent literature and thus poses a risk of a data leak.

2.4 Exploratory Data Analysis

Discuss missing values, outliers, normalization, feature encoding, or transformations.

2.4.1 Missing Values

Among collected data and chosen features, only a particular group of drug features present missing values as shown in Figure A4. To find out what it is linked to, we created an indicator variable for missingness for this group of features and built a correlation heatmap to visualize missingness patterns (Figure A5). Indeed, this group of chemical descriptors (such as number of aromatic rings or lipophylicity) tend to be absent together and they are associated with biotherapeutics marker. Biotherapeutics are drugs derived from living organisms rather than being chemically synthesized. Most of them are antibodies or other proteins - large biopolymers for which this group of chemical descriptors is not relevant. Indeed, manual inspection of drugs with missing values confirms this (e.g. with insulin among such drugs). So we decided to consider only small molecule drugs since chemical description of biopolymers requires another approaches, imputing those missing chemical descriptors would be a mistake. After the removal of those cases, data is ready for merging, the result containing all information on each drug-disease indication. Table 2 provides a summary of data dimensions and label counts.

Table 2: Summary of data dimensions and label counts.

Dataset	Rows	Columns
Drugs	525	181
Diseases	149	79
Trials	14316	12
Indications	2482	12
Name embeddings	489939	5
Molecular fingerprints	1029	101
Merged data	890	252
Merged data success counts	False: 637, True: 253	

2.4.2 Univariate and bivariate analysis

Inspection of feature distributions across label values of variables not related to molecular pathways or fingerprints (Figure A6) shows approximately bell-shaped but sometimes multimodal or slightly skewed distributions. Nevertheless, we don't find log transformation of the features necessary here. As for disease ontologies or some drug descriptors, there are some outliers, but it may be a part of normal biological or chemical variability so we will analyze potential outliers with dimension reduction techniques. We can also notice that feature distributions are rather similar between success and failure groups. It's expected since unfortunately such basic descriptors alone cannot discriminate success cases. However, we can notice that engineered name and path similarity features tend to be higher for success groups, this indicates that those variables might be promising predictors combined with other variables, although name similarity was excluded from further analysis since was considered as spoiler.

On a correlation heatmap for the whole merged dataset (Figure A7) we can notice that some chemical descriptors with molecular fingerprints and drug pathway embeddings, disease ontology types with disease pathway embeddings form, as expected, clusters with strong or moderate associations. When we focus on more interpretable features (without structural or mechanistic embeddings, Figure A8), we can explain some of them. For example number of heavy atoms is associated with molecular weight. Number hydrogen bond acceptors and donors (HBA and HBD) are supposed to be associated with higher hydrophilicity of a molecule and it indeed correlates positively with polar surface area (PSA) and negatively with octanol-water partition coefficient (AlogP). Number of rule of 5 violation is a feature based on some of those chemical descriptors so it is, as expected, associated with them. Interestingly, number of aromatic rings correlates negatively with the indicator that the drug was obtained from the natural source (NP-likeness score). Apparently, purely synthetic drugs contain more aromatic rings.

As for disease ontology, we can for example see that immune diseases labelled as hemic or lymphatic tend to be neoplastic as well, it makes sense since most of oncological conditions concerning the immune system are indeed various types of leukemias and lymphomas. Another example is positive association between urogenital diseases and infections.

2.4.3 Dimension Reduction and Outlier Analysis

When we perform PCA on centered and scaled merged data (Figure A9) we can first notice that first two principal components explain only 10 % of total variance, and most of the variance is distributed among a large number of components as can be seen on scree plot in Figure A10. It can mean that no linear combination of these features form a single or several dominant axes of data variability. This suggest that the feature space is intrinsically high dimensional and associations between variables are likely non-linear which is expected for such a complex system. Furthermore, a lot of features were explicitly designed to be uncorrelated (principal components for molecular fingerprints and pathways embeddings). We can however cautiously take a look at the data projection on first two principal components and notice that the major variation is associated with disease heterogeneity, since the most prominent arrows on correlation plot (Figure **??**) associated with first principal components are disease category indicators and disease pathway embeddings, for example neoplastic blood disorders as opposed to respiratory tract diseases.

The second principal component is rather associated with drug related variability, in particular we can notice a group of potential outliers for high values of firs and second principal component. Their closer inspection reveals that these indications involve three drugs sirolimus, everolimus and ridaforolimus. These drugs turn out to be antineoplastic and immunosuppressor drugs with the same structural code - large macrolide ring (Figure A12). We considered it as a part of normal drug variation that cannot be attributed to a mistake or poor data filtering and thus we decided to not exclude them.

Although PCA is unsupervised method, when we label the data a posteriori, we can see a slight separation of success and fail cases along the diagonal between the first two principal components. Interestingly enough, a Quantitative Estimate of Druglikeness (QED) contributes to both principal components and is aligned with samples labelled as success. This suggests that the first independent directions of greatest variability in the data capture in part the difference in indication success. We could also speculate that it might be linked to different success rates for various disease types, for example clinical studies for neoplastic blood diseases might have fewer successes because of smaller success rate or may be because they are more often studied.

Non linear dimension reduction methods such as Isomap, UMAP and t-SNE (Figures A13 and A14 and A15) result in a rather large number of small clusters. Apparently, those clusters correspond to the indications involving same of very similar drugs or diseases.

2.4.4 Supervised Dimension Reduction

By performing supervised dimension reduction models - LDA and PLS-DA (Figures A16 and A17) we can see that the label distributions along LDA axis or in projection to first two components of PLS-DA, are, although overlapping, are rather well separated

2.5 Graph Representation

We need to acknowledge that the chosen approach for binary classification of drug-disease indications, although is valid, doesn't model the true structure of the problem.

Indeed, the data is inherently structured and a bipartite drug-disease graph with a connecting layer comprised of associated molecular pathways. And most of advanced research in the field of drug discovery involve working with graphs and Graph Neural Networks directly rather than engineering features as we did [Keramida et al., 2025]. To this end, we organized the same data in the form of `HeteroData` graph with Pytorch Geometric module. Although we think that the richer information about connections on a molecular / mechanistic levels as well as more observations are needed to build a graph suitable for the robust model training.

```
HeteroData(
  drug={ x=[525, 127] },
  disease={ x=[149, 25] },
  pathway={ x=[1487, 1] },
  (drug, interacts, pathway)={ edge_index=[2, 17338] },
  (disease, associated_with, pathway)={ edge_index=[2, 12064] },
  (drug, tryed_against, disease)={
    edge_index=[2, 890],
    edge_label=[890],
  }
)
```

3 Methods

3.1 Model Overview

Briefly describe the models used. Include general principles (e.g., linear models, random forests, neural networks).

Since the prediction target is a binary success/failure for given drug-disease indication, we evaluated several supervised models for binary classification, including Logistic Regression, k-Nearest Neighbors (kNN), Support Vector Machines (SVM), Random Forest, Gradient Boosting, and XGBoost.

Logistic regression estimates probability of success using sigmoid-transformed linear combinations of the input features. Its biggest advantage is high interoperability since feature waits indicate their importance, however it is limited to linearly separable problems.

k-Nearest Neighbors is instance based model, making predictions by looking at labels of the closest points in the feature space. Therefore, there is no actual training phase and the assumption behind this model is that similar observations tend to have the same label. Nevertheless it is very sensitive to noise and the curse of dimensionality, as well as poorly interpretable (only by samples that voted).

Support Vector Machines are fitting a decision boundary, maximizing the margin between classes. Although it may be similar to regularized linear regression, the usage of kernel functions allows fitting of non-linear decision boundaries.

Tree-based ensemble learning models leverage hierarchical decision rules and combine prediction of multiple models. A random forest aggregates multiple decision trees based on random subsets of variables and observations, this decreases the overfitting so characteristic of decision trees. Gradient boosting and XGBoost instead build trees sequentially, where each new tree learns errors of the previous ones. XGboost is an optimized version of Gradient Boosting, but has more parameters to tune.

As for graph-based approach, we chose to use a Heterogenous Graph Neural Network (GNN) link classifier from PyTorch Geometric module. Instead of learning patterns based on tabular features only, it learns representations by propagating information across connected nodes and edges, enabling the model to capture more complex relationships. This approach avoids manual feature engineering, but requires a large number of samples and rich informative connections in a graph, and, as other Artificial Neural Networks, prone to overfitting.

3.2 Implementation Details

Mention important parameters, software/libraries used.

All numeric features are centered and scaled. No imputation was needed since observations with missing values corresponding to biopolymer drugs were excluded.

Most of the mentioned models are imported from `sklearn` module, except for XGBoost imported from `xgboost` and GNN from `torch_geometric` modules. In order to perform hyperparameter tuning, we used a grid-search cross validation `GridSearchCV` with search spaces indicated in Table 3.

Since most of the clinical trials and thus corresponding indications usually fail, the classes are imbalanced with failures being several times more abundant (around 3 times for our data). So we used options for class balancing where it was possible (all models except for kNN and Gradient Boosting).

Table 3: Hyperparameter search spaces for each classifier.

Model	**Hyperparameters**
k-Nearest Neighbors	`n_neighbors` $\in \{5, 10, 20, 30\}$, `weights` $\in \{\text{uniform}, \text{distance}\}$, `p` $\in \{1, 2\}$
Logistic Regression	`C` $\in \{0.01, 0.1, 1, 10, 100\}$, `l1_ratio` $\in \{0, 0.25, 0.5, 0.75, 1\}$
SVM (linear)	`C` $\in \{0.01, 0.1, 1, 10\}$
SVM (polynomial)	`C` $\in \{0.1, 1, 10\}$, `degree` $\in \{2, 3, 4\}$, `gamma` $\in \{\text{scale}, \text{auto}\}$
SVM (RBF)	`C` $\in \{0.1, 1, 10\}$, `gamma` $\in \{\text{scale}, \text{auto}\}$
Random Forest	`n_estimators` $\in \{100, 300, 500\}$, `max_depth` $\in \{\text{None}, 10, 20\}$, `min_samples_split` $\in \{2, 5\}$, `min_samples_leaf` $\in \{1, 2\}$
Gradient Boosting	`n_estimators` $\in \{100, 300\}$, `max_depth` $\in \{3, 5\}$, `learning_rate` $\in \{0.01, 0.05, 0.1\}$, `subsample` $\in \{0.8, 1.0\}$
XGBoost	`n_estimators` $\in \{100, 300\}$, `max_depth` $\in \{3, 5\}$, `learning_rate` $\in \{0.01, 0.05, 0.1\}$, `subsample` $\in \{0.8, 1.0\}$, `colsample_bytree` $\in \{0.8, 1.0\}$, `gamma` $\in \{0, 0.1\}$

3.3 Evaluation Strategy

Drug-disease indication success prediction requires avoiding temporal data leak - in order to simulate fairly the real application, we need to use older research results to predict success of new indications. Thus, we performed a train-test split based on on start date of the first clinical trial for each indication. Taking 15% for test set results in temporal splitting by August 2014.

The training set is further used for hyperparameter tuning using grid search with 5-fold cross-validation and model selection based on cross-validation performances.

Since we deal with imbalanced classes, we must not use accuracy as evaluation metric since it will underemphasize incorrect prediction of minority (success) class. We thus used F1 score (harmonic mean of precision and recall) for model hyperparameter tuning, selection and evaluation, as well as precision-recall curves instead of ROC curves for analyzing how clarification result varies based on decision threshold (it works only for models yielding class probabilities).

4 Results

4.1 Model Comparison

Compare models quantitatively (accuracy, RMSE, AUC, etc.) and qualitatively (strengths/weaknesses).

Best estimators after hyperparameter tuning are presented in Table 4 and the corresponding hyperparameters in Table A2. Based on mean cross-validation scores, we can say that all models perform relatively similarly having mean F1 score between 0.6 and 0.65. The best one is XGBoost and the worst one is kNN, what is expected given high dimensionality of the problem.

Table 4: Model performance after hyperparameter tuning (5-fold cross validation).

Model	**Mean F1 score**	**Std F1 score**
XGBoost	0.656383	0.038165
Logistic Regression	0.619974	0.037465
Support Vector Machine	0.619910	0.033970
Gradient Boost	0.609287	0.092787
Random Forest	0.601969	0.044340
k-Nearest Neighbors	0.591973	0.052311

We will use then XGBoost for final evaluation on test set and we obtain precision 0.658, recall 0.595 and F1 score 0.625 (Figure A18).

Precision-recall curve (Figure A19) presents rather smooth descent from 1 to chance level (success prevalence of 0.313). It means that there are no abrupt drops in precision as model becomes less strict (higher recall) and that model performance is rather stable. Area under the ROC curve is 0.824 (Figure A20), but it is a less reliable metric in case of class imbalance.

We performed as well training of Heterogenous GNN and obtained accuracy 0.612, precision 0.429, recall 0.714 and F1 score 0.535 (Figure A21). Unlike the other machine learning models, the GNN was evaluated independently for two reasons. First, GNN implementation is rather exploratory in this project and we cannot yet ensure that all GNN learning parameters were correctly configured. Second, differences in model architecture made made direct comparison with other models in the common cross-validation setting non-trivial. Therefore, GNN results should be interpreted as complementary and preliminary.

4.2 Feature Analysis

Analyze the impact of key features on model performance (e.g., feature importance, correlation).

Since XGBoost model is not interpretable by design, we need to use post-hoc explanation strategy. We chose to use a permutation feature importance estimation (still using F1 score) and partial dependence analysis from `sklearn` module (Figure A22). Permutation of pathway similarity feature results in 15% decrease in F1 score, around 3.5% for number of indications for a disease (proxy of disease "popularity") and 6-th component in disease pathway embedding space. Other features, including various disease and drug pathway embeddings a nd molecular fingerprints present less than 3.5% decrease in F1 score.

Partial dependence plot (Figure A23) shows that the steepest increase in success prediction happens at around 0.25 for pathway similarity and 90 for number of indications for a disease. Two-dimensional partial dependence plot for pathway similarity and number of indications (Figure A24) shows that the model needs simultaneously high pathway similarity and large number of indications to be sure in success prediction.

4.3 Failure and Success Cases

Highlight scenarios where the models perform well or poorly. Include illustrative figures or tables.

We added to the test set features indicating whether each observation is true positive (TP), true negative (TN), false positive (FP) or false negative (FN) and just correct (TP or TN).

We can analyze Pearson correlations of those confusion estimates with predictor features (being cautious with interpretation since these introduced variables are binary labels, so it must be interpreted as point-biserial correlation) (Figures A25 and A27) as well as distributions of predictors in correctly classified and misclassified cases (Figures A26 and A28). We can see that there is no major difference between those cases. The distributions for some variables present a slight bias, but their interpretation will be speculative.

5 Discussion

Interpret the results. Relate findings to your original objectives. Discuss limitations and possible improvements.

We observe best tuned model, XGBoost has moderate performance on the test data, with having slightly better precision (0.658) than recall (0.595). It means for this test dataset when the model has more false negatives than false positives. Since the ultimate aim of the model application is to prioritize new drug-disease indications, better precision yields makes results that are predicted as positive more reliable, but worse recall means that some combination may be (potentially systematically and unfairly) ignored. Other models performed slightly worse, especially kNN, this suggests that those models didn't capture well non-linear patterns in noisy multidimensional data or failed to handle class imbalance.

Heterogenous GNN was trained and tested separately and yielded a large number of false positives resulting in low precision of 0.429. This poor performance can be explained by poor handling of class imbalance on our side, since we used binary cross entropy loss with positive weights equal to number of times success labels are less prevalent and it might have been to stringent to the model. We should further analyze the correct ways to handle class imbalance in GNN including proper hyperparameter and classification threshold tuning. But considering other potential explanations for a poor performance, we could argue that the size of training data was not enough to properly train such a complex model. We also need to discover ways to handle potential overfitting issues with GNN.

Two most important features identified by permutation method are pathway similarity and number of indications for a disease, and both these variables need to be high for assignment of large success probability. It is expected to see pathway similarity being predictive since this metric captures shared molecular mechanisms behind action of a drug and pathogenesis of a disease. It is also unfortunately expected to see number of indications being predictive of success since this feature reflects how intensively researched is the disease, this constitutes a bias towards well-studied diseases and potentially neglecting rare or poorly studied ones. However, analysis of feature distributions for correctly classified and misclassified cases didn't reveal any prominent bias. So probably the importance of number of indications reflects a bias in clinical research itself rather than a bias in the model.

We can also identify

... biases: publication, data availability, rare diseases, a lot of unknowns, obvious negatives are not tried ...

6 Conclusion

Summarize your work, key findings, and potential future directions.

Although the evaluation results show only moderate prediction success and biases, it is still surprising that such a simple metric (number of shared pathways) combined with simple chemical and molecular descriptors can be predictive of a success, unless we missed some other biases.

... improvements: PU, explore more Graph models, more targets, consider directionality, more diseases, better labels, consider biopolymers, smartly include obvious negatives....

References

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A Appendix

A.1 Supplementary Tables

Table A1: Description of features retained for analysis.

Feature	Type	Description
Drug		
drug_id	str	Unique ChEMBL identifier
chemical_probe	bool	Well-characterized and selective, typically used in biological research
dosed_ingredient	bool	Active ingredient that has been dosed in humans
inorganic_flag	bool	Inorganic molecule
natural_product	bool	Derived from a natural source (not fully synthetic)
oral	bool	Known to be administered orally
orphan	bool	Intended for use against a rare condition
parenteral	bool	Known to be administered parenterally
prodrug	bool	Prodrug, activated after administration
therapeutic_flag	bool	Has a therapeutic application, opposed to e.g. imaging
topical	bool	Known to be administered topically
veterinary	bool	Has veterinary application as well
chirality	cat	achiral,single enantiomer, mixture,nan
alogp	float	Octanol-water partition coefficient, lipophilicity indicator
aromatic_rings	int	Number of aromatic rings
full_mwt	float	Molecular weight of the full compound including any salts
hba	float	Number of hydrogen bond acceptors
hbd	float	Number of hydrogen bond donors
heavy_atoms	int	Number of heavy (non-hydrogen) atoms
mw_freebase	float	Molecular weight of parent compound
np_likeness_score	float	Natural product likeness score [Ertl et al., 2008]
num_ro5_violations	int	Violations of Lipinski's rule-of-five, using HBA and HBD definitions
psa	float	Polar surface area
qed_weighted	float	Weighted quantitative estimate of drug likeness [Bickerton et al., 2012]
ro3_pass	int	Passes the rule-of-three (mw $<$ 300, logP $<$ 3 etc)
rtb	int	Number of rotatable bonds
FP_#	float	First PCs in Morgan molecular fingerprint space
drug_path_#	float	First PCs in TF-IDF space derived from pathway lists
Disease		
disease_id	str	Unique EFO identifier
ontology_depth	int	Depth in disease classification tree
nb_descendants	int	Number of descendants in disease classifications tree
nb_indications	int	Number of drug indications
cat_#	bool	Ontological category other than immune system disorders
disease_path_#	float	First PCs in TF-IDF space derived from pathway lists
Drug–Disease		
success	bool	Consensus success or failure
path_similarity	float	Jaccard similarity of pathway lists
name_similarity	float	Cosine similarity of name embeddings in PubMedBERT (later excluded due to data leak)

Table A2: Best estimator parameters after hyperparameter tuning.

Model	Best parameters
XGBoost	colsample_bytree = 0.8,gamma = 0,learning_rate = 0.01,max_depth = 5,n_estimators = 300,subsample = 1.0
Logistic Regression	C = 0.1,l1_ratio = 0.25
Support Vector Machine	C = 1,degree = 2,gamma = scale,kernel = poly
Gradient Boost	learning_rate = 0.05,max_depth = 3,n_estimators = 100,subsample = 0.8
Random Forest	max_depth = 10,min_samples_leaf = 2,min_samples_split = 2,n_estimators = 100
k-Nearest Neighbors	n_neighbors = 5,p = 2,weights = distance

A.2 Supplementary Figures

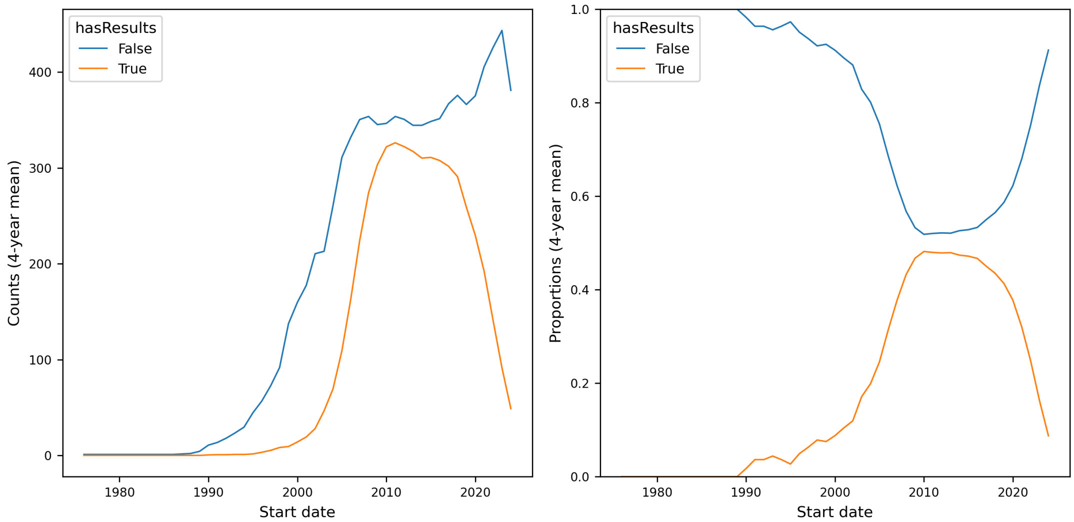

Figure A1: Counts and proportions of clinical trials with published results over time.

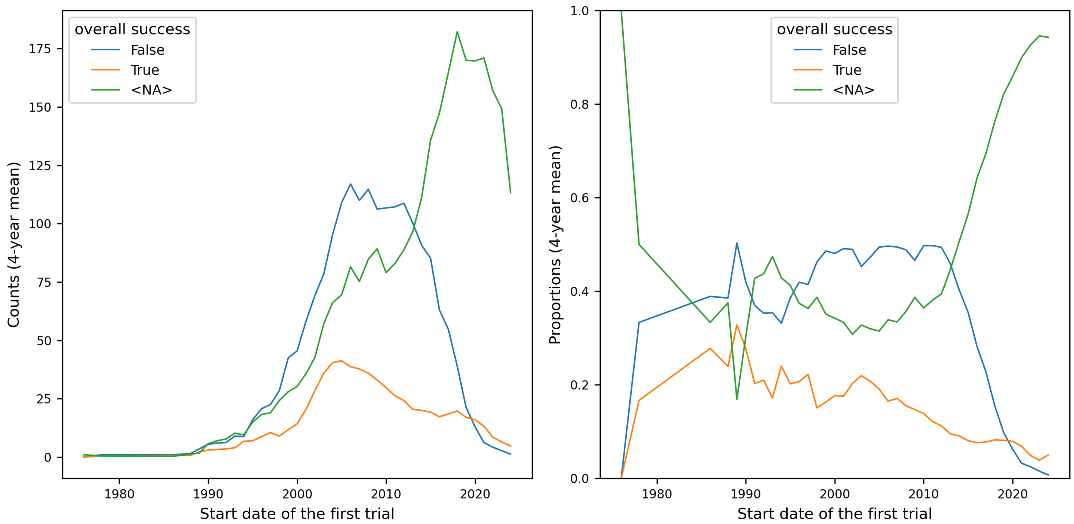

Figure A2: Counts and proportions of drug-disease indications by success label over time.

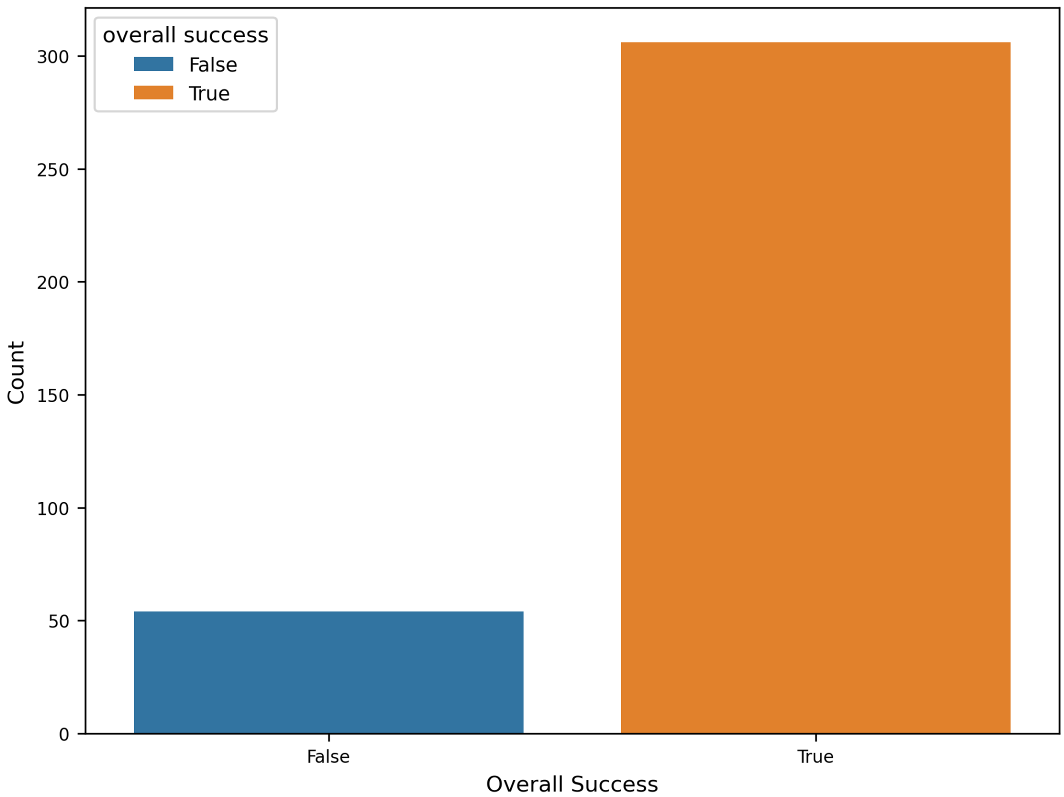

Figure A3: Distribution of success labels for drug–disease indications with missing start dates.

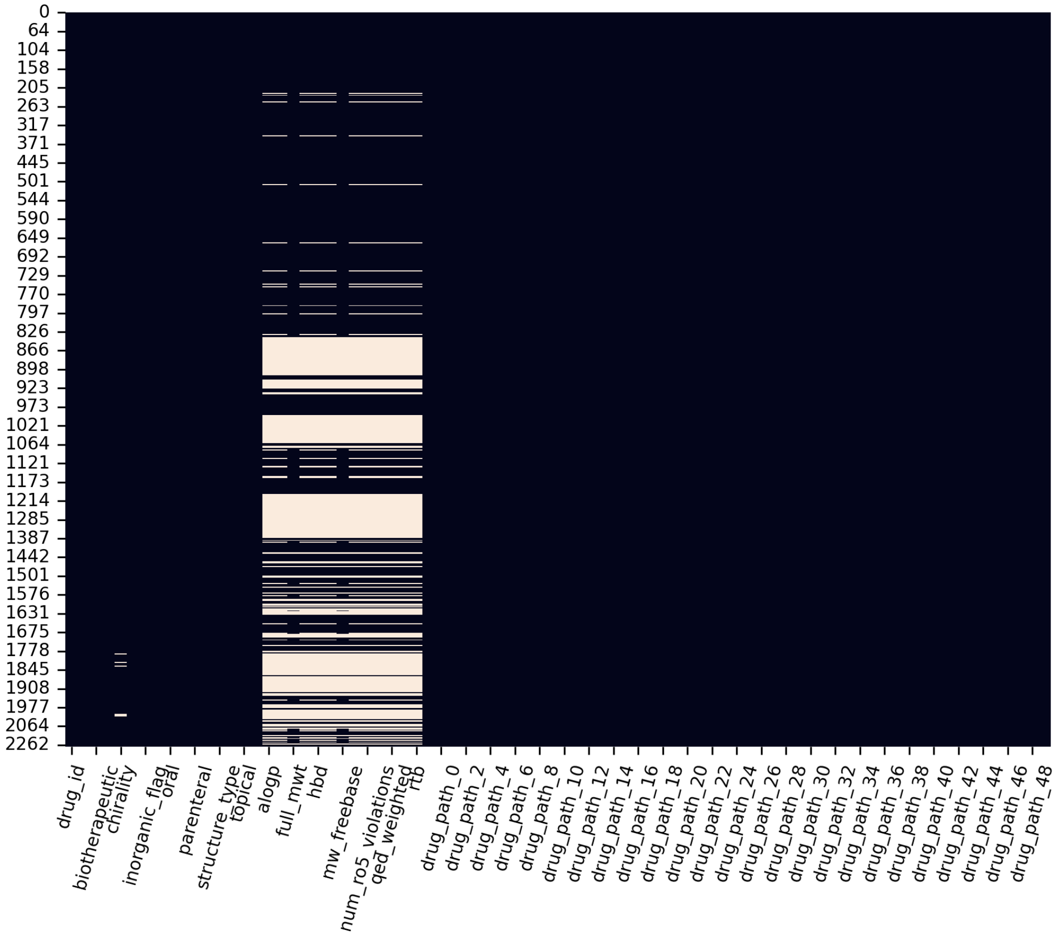

Figure A4: Missing values heatmap in the drugs dataset.

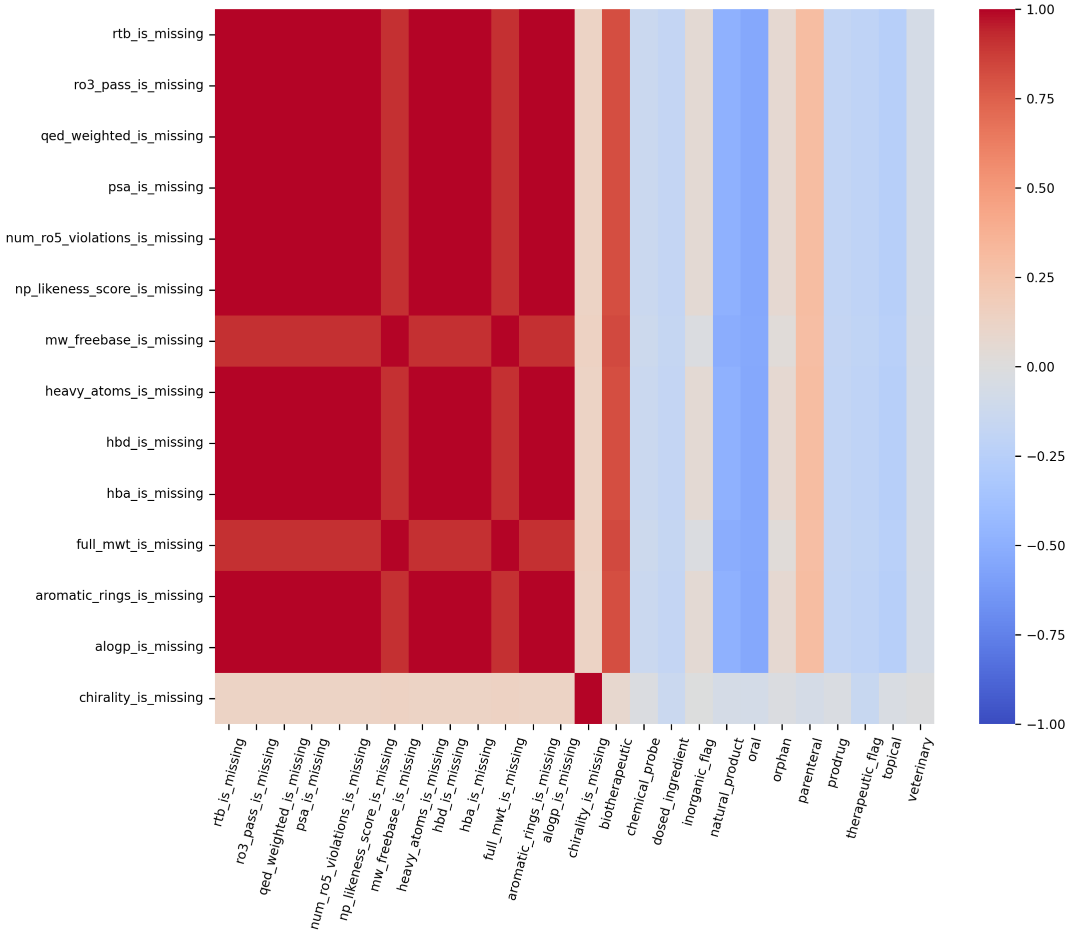

Figure A5: Missingness analysis of drug descriptors with correlation heatmap.

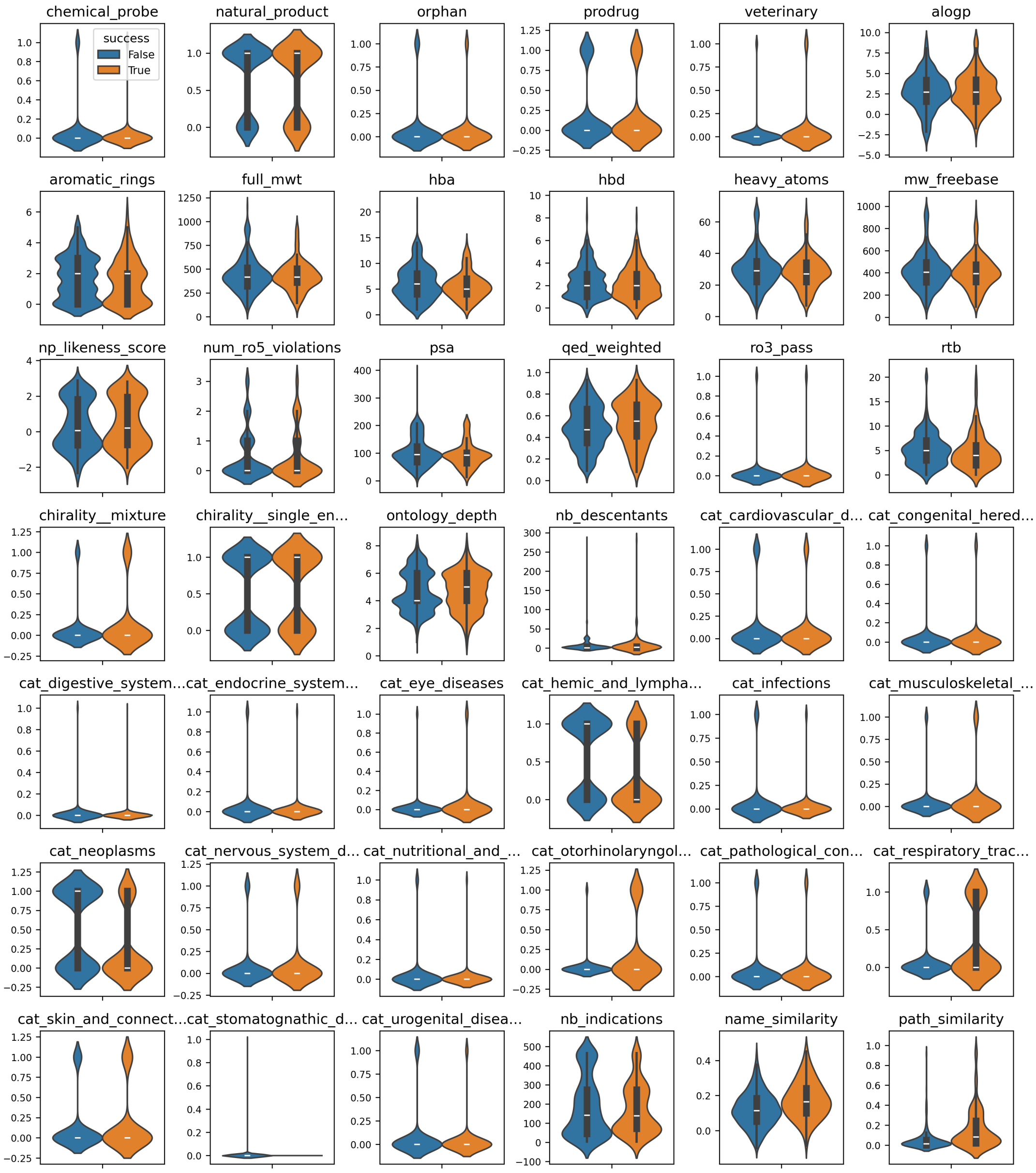

Figure A6: Feature distributions by success label.

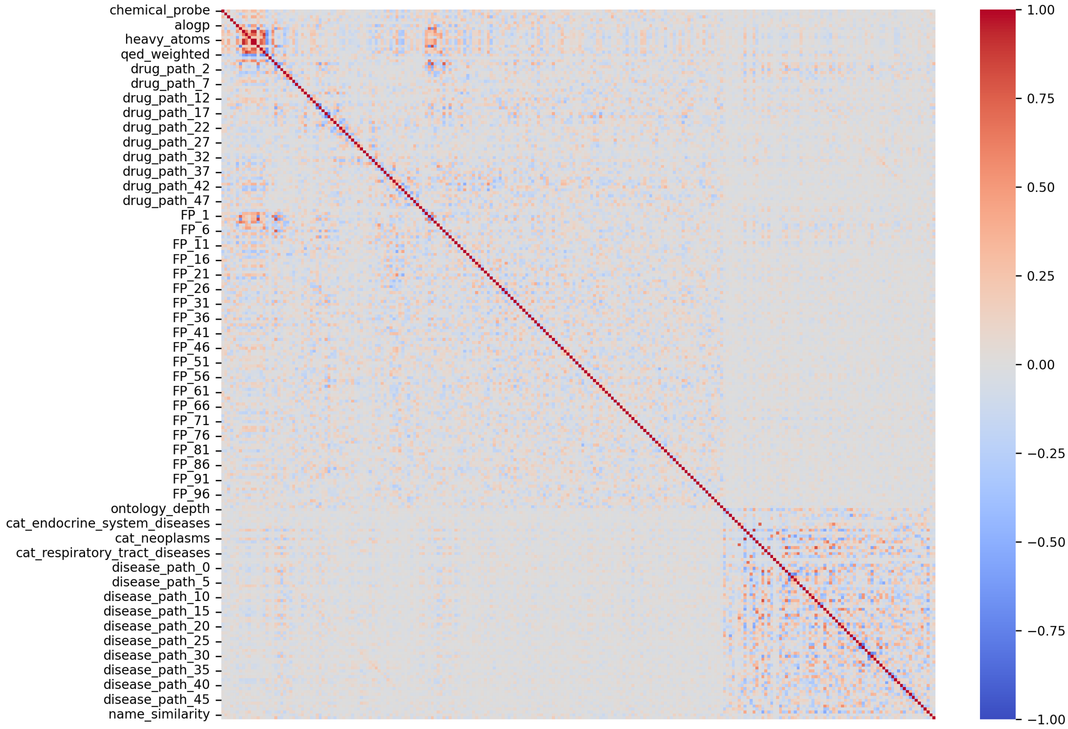

Figure A7: Correlation heatmap of all features.

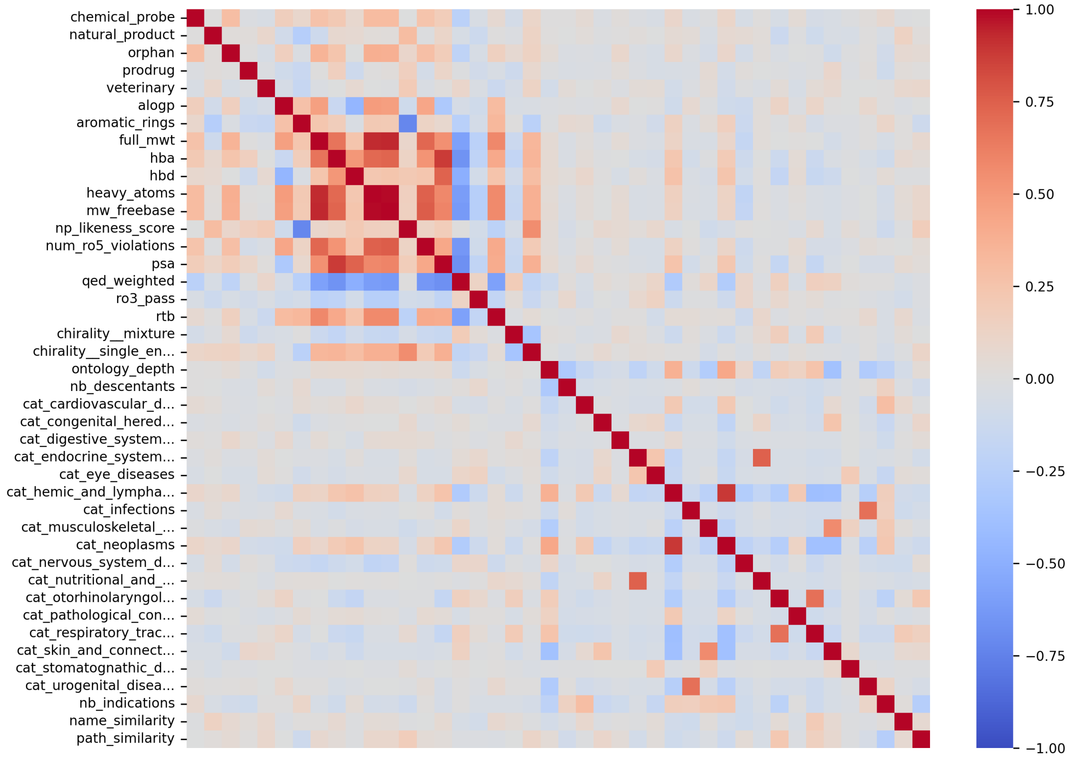

Figure A8: Focused view of the correlation heatmap on interpretable features.

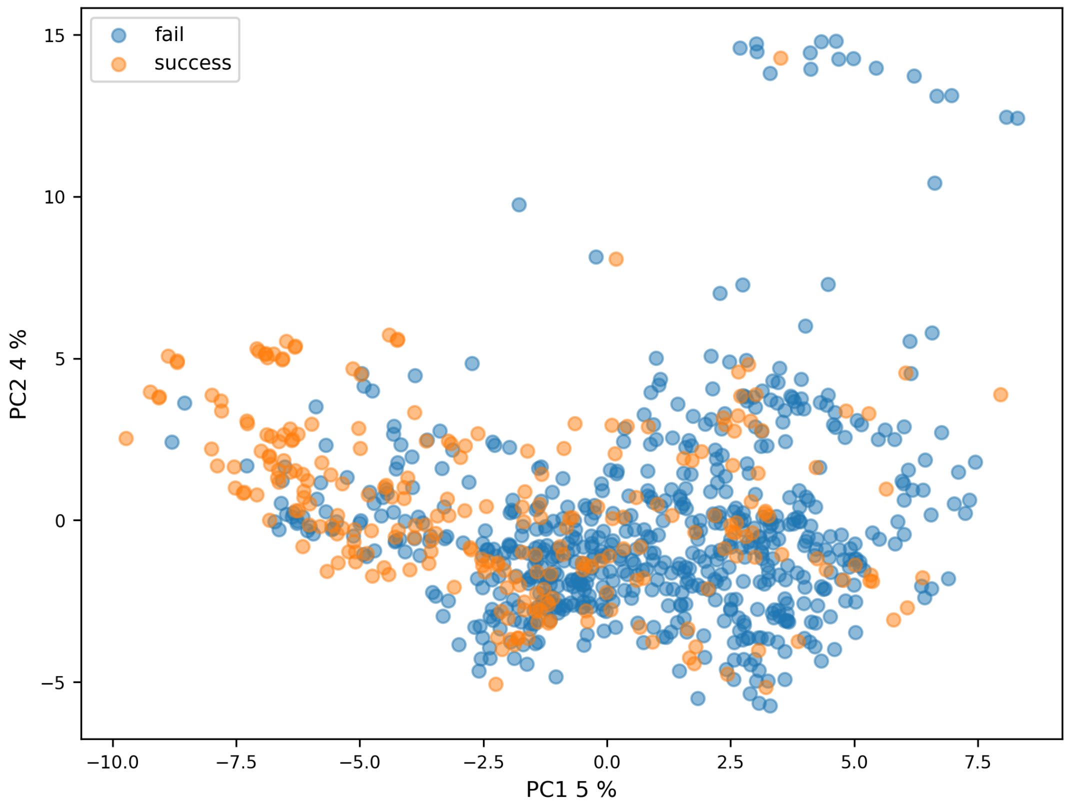

Figure A9: PCA projection of samples onto the first two principal components.

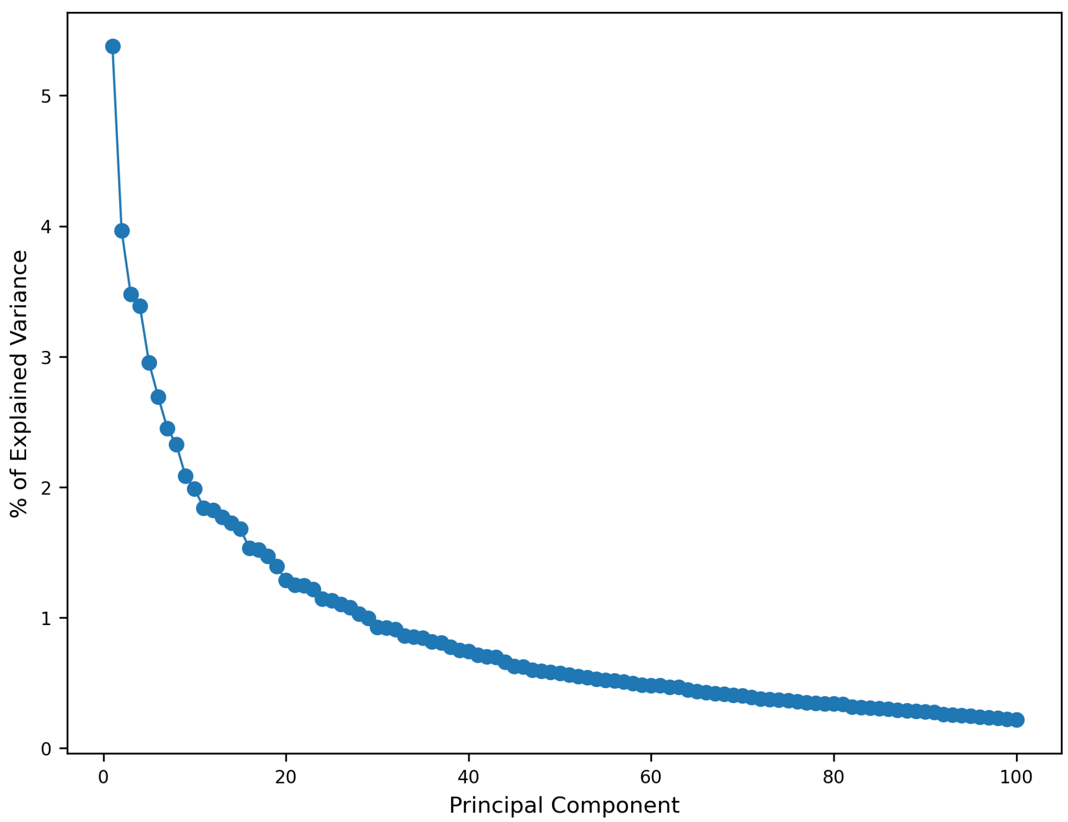

Figure A10: Scree plot of explained variance for PCA components.

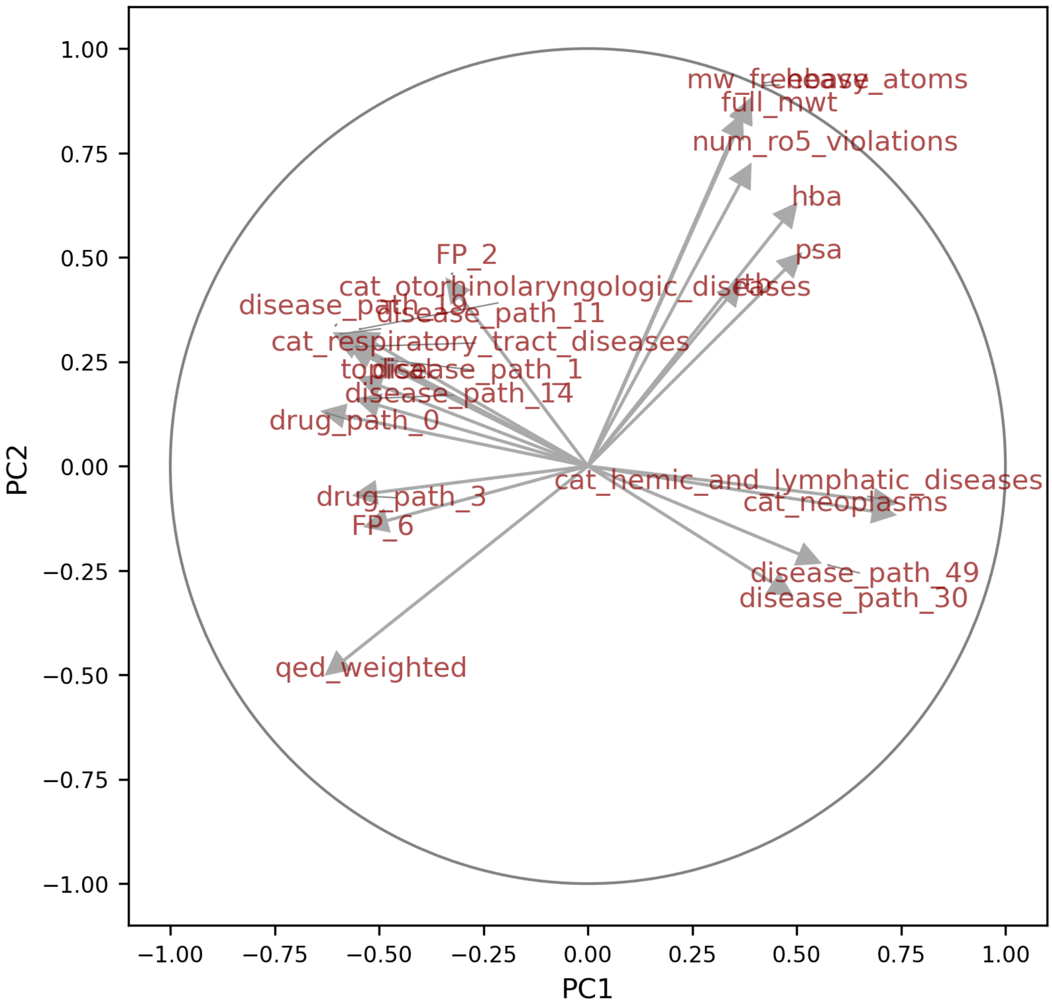

Figure A11: PCA correlation circle showing feature contributions to the first two principal components.

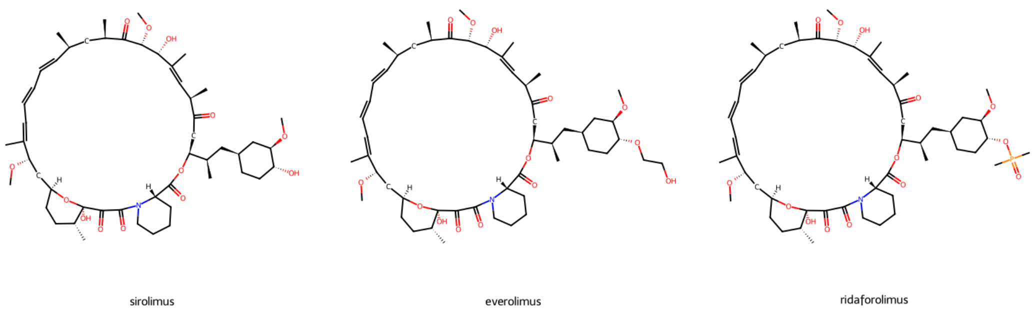

Figure A12: Structures of drugs corresponding to outlier indications.

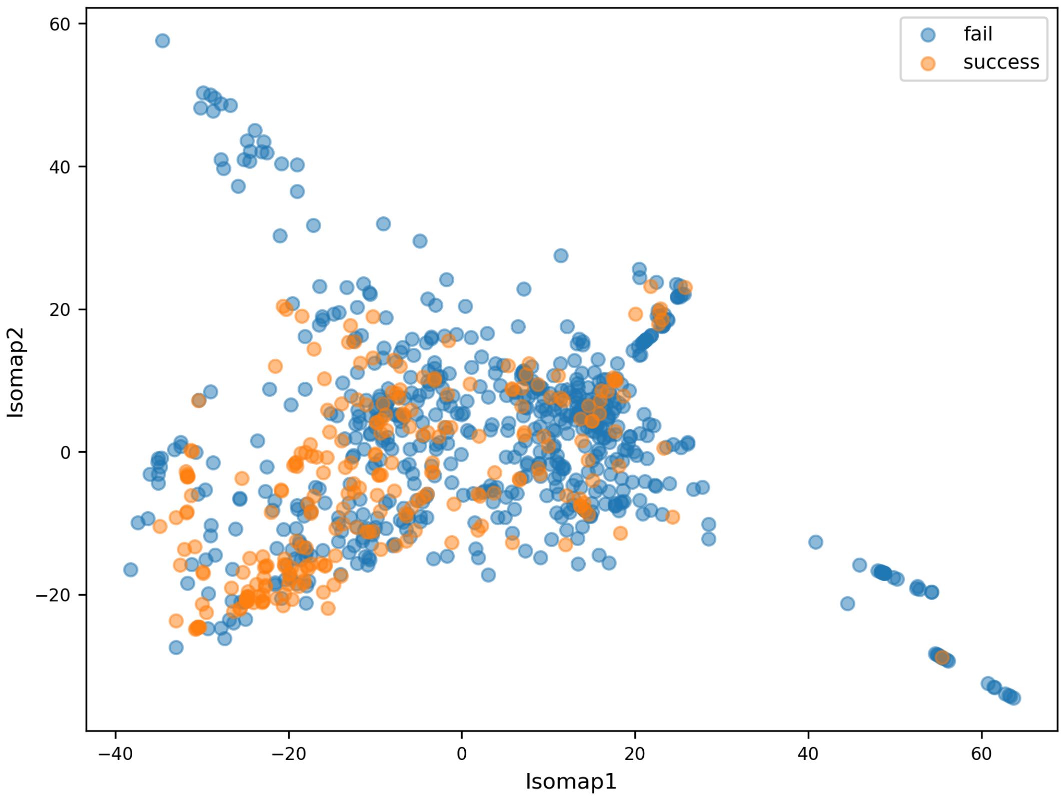

Figure A13: Isomap projection of samples in the feature space using 10 nearest neighbors.

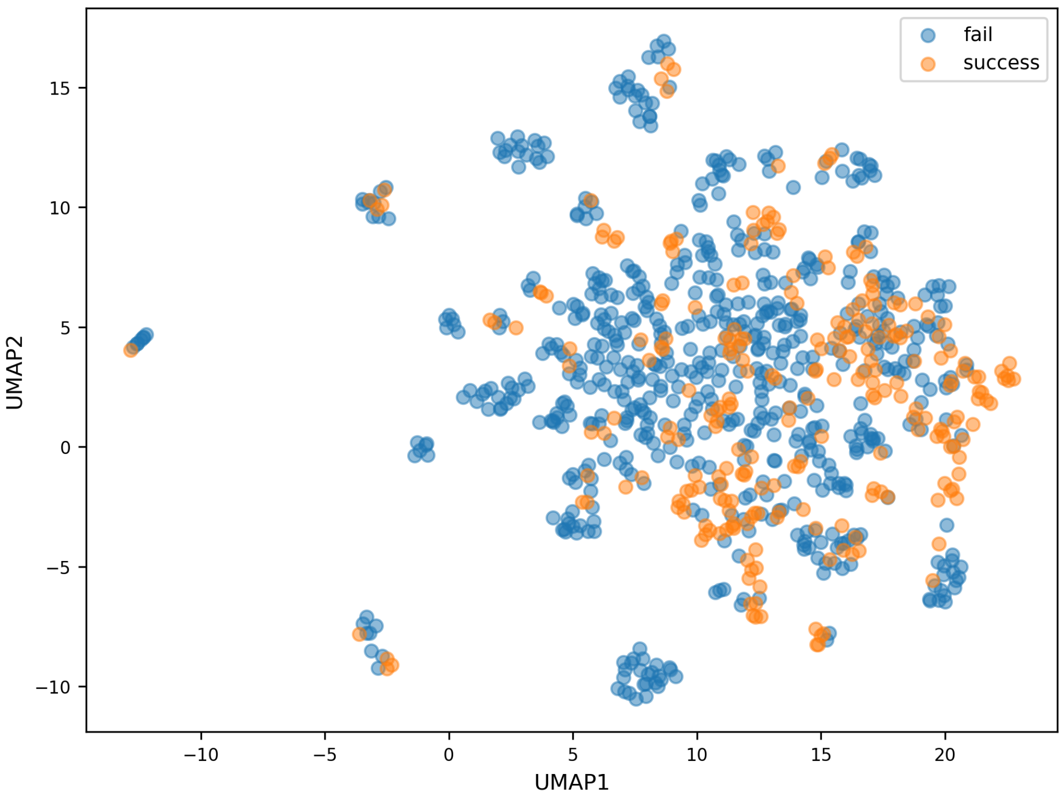

Figure A14: UMAP projection of samples in the feature space using 7 nearest neighbors and a minimal distance of 1.

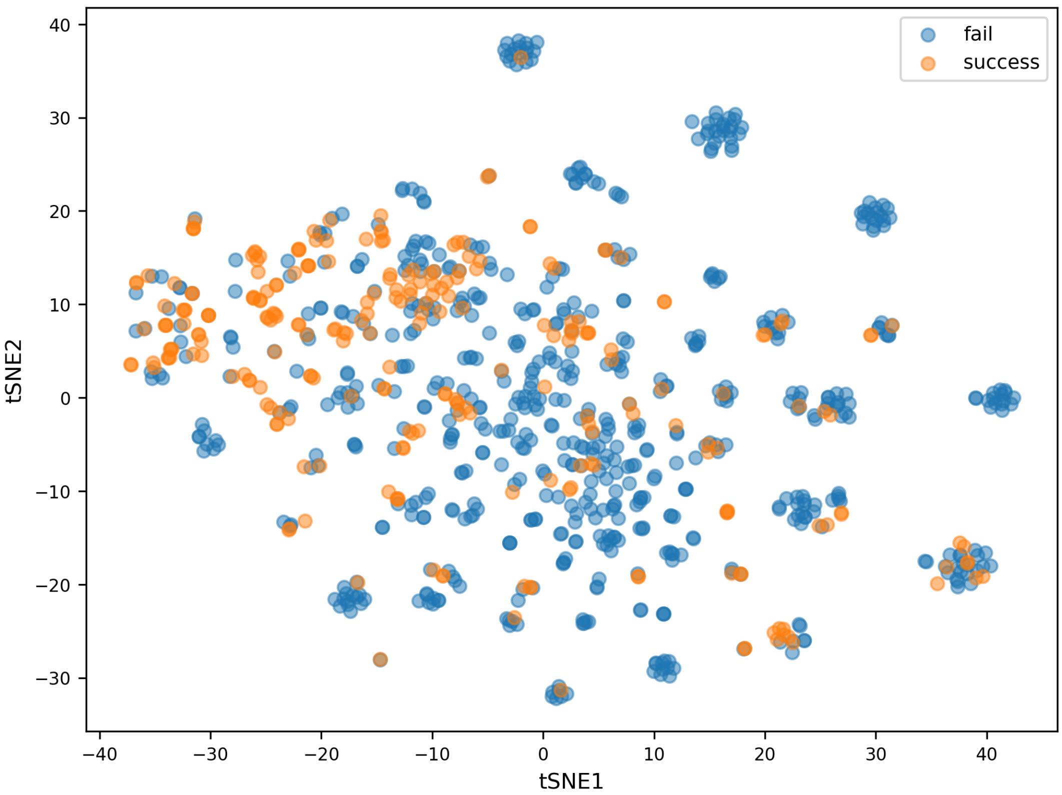

Figure A15: t-SNE projection of samples in the feature space using a perplexity of 40.

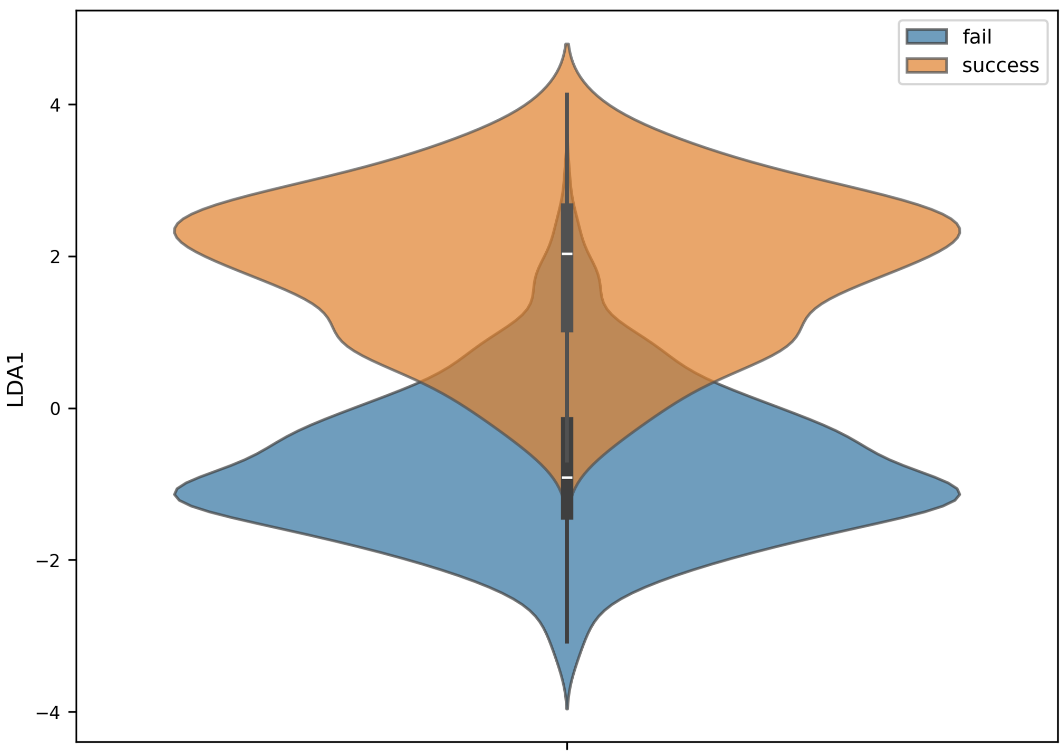

Figure A16: Linear Discriminant Analysis (LDA) projection highlighting class separation.

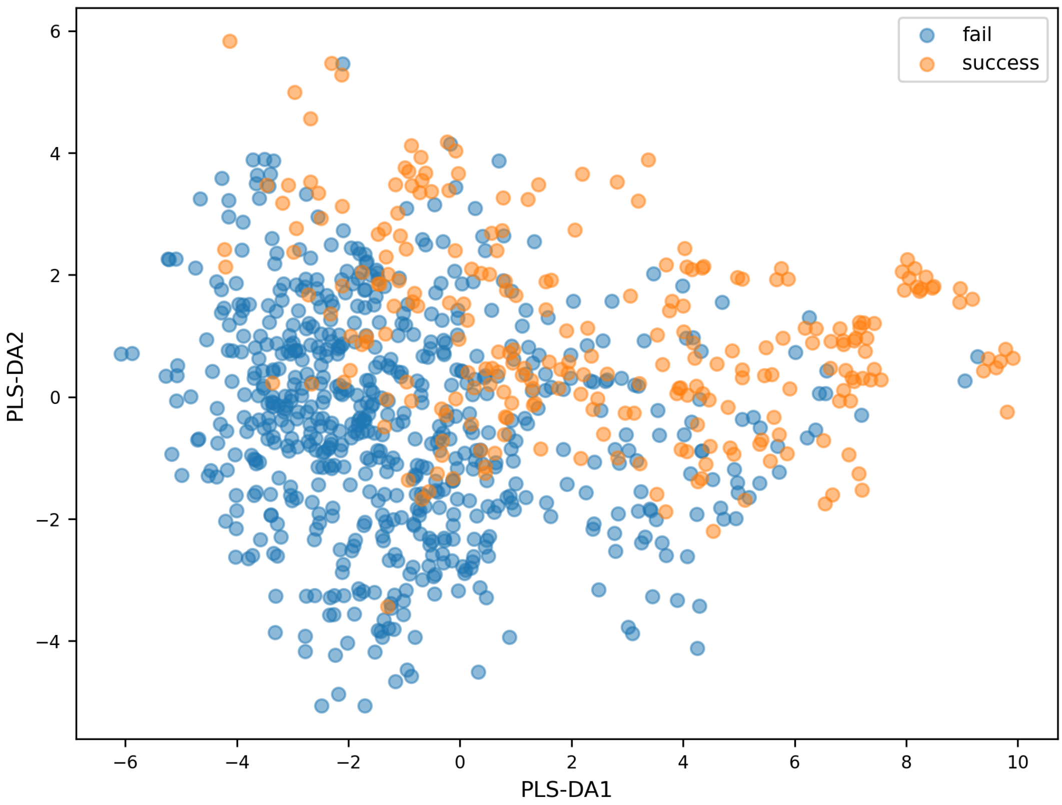

Figure A17: Partial Least Squares Discriminant Analysis (PLS-DA) projection of samples highlighting class separation.

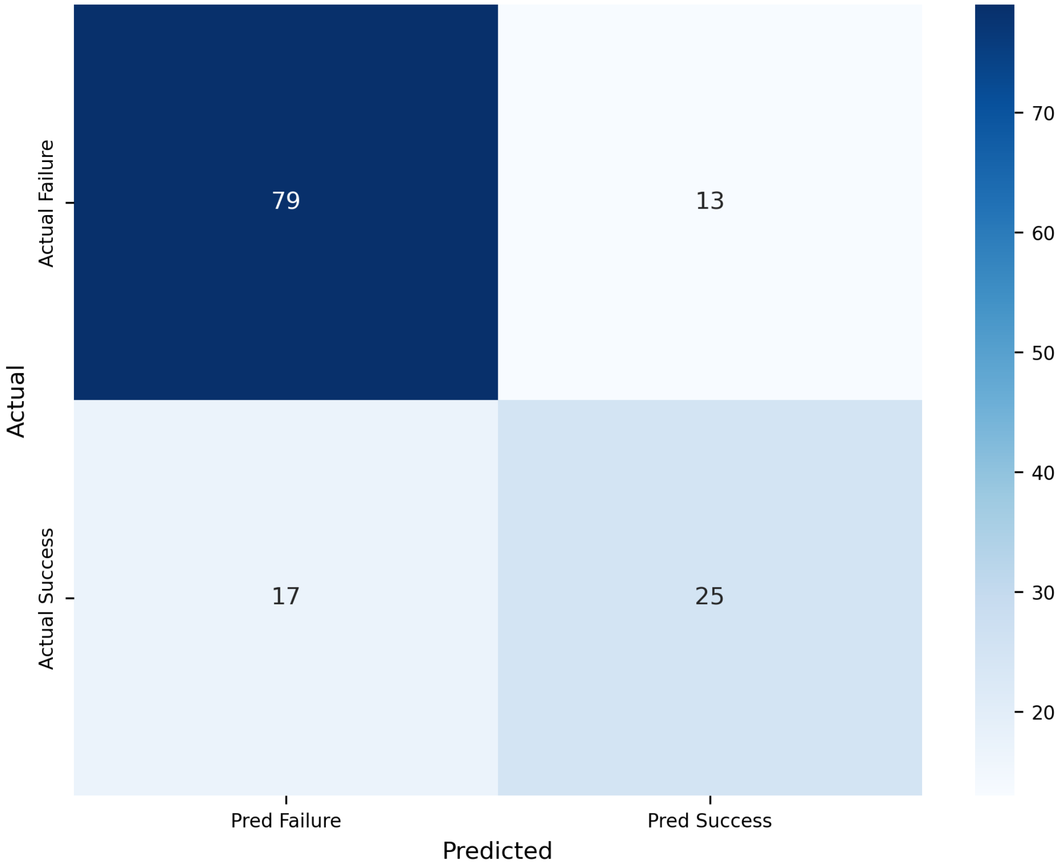

Figure A18: Confusion matrix for XGBoost on the test dataset.

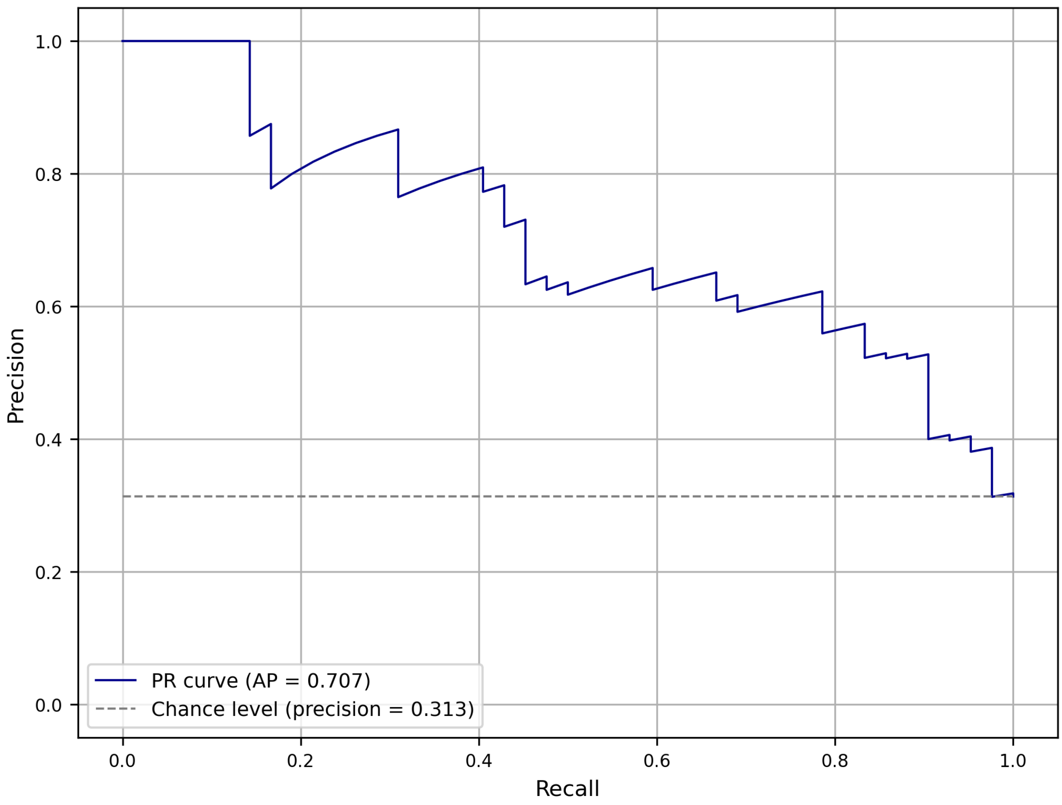

Figure A19: Precision-recall curve for XGBoost on the test dataset.

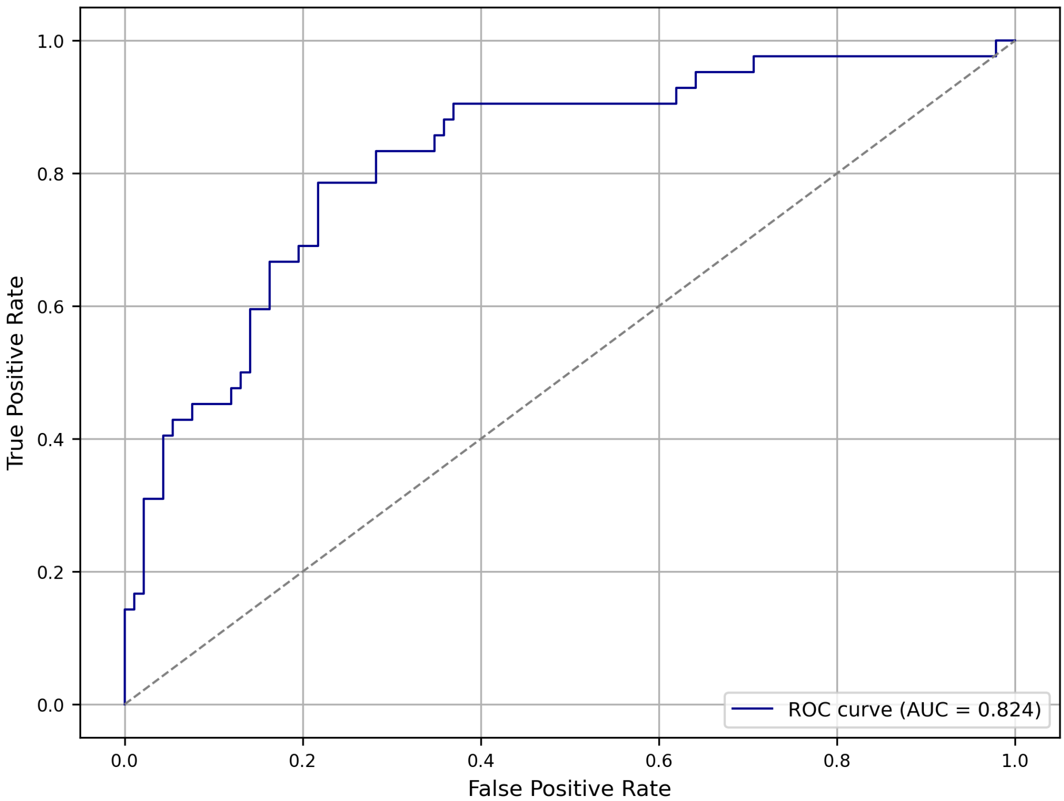

Figure A20: ROC curve for XGBoost on the test dataset.

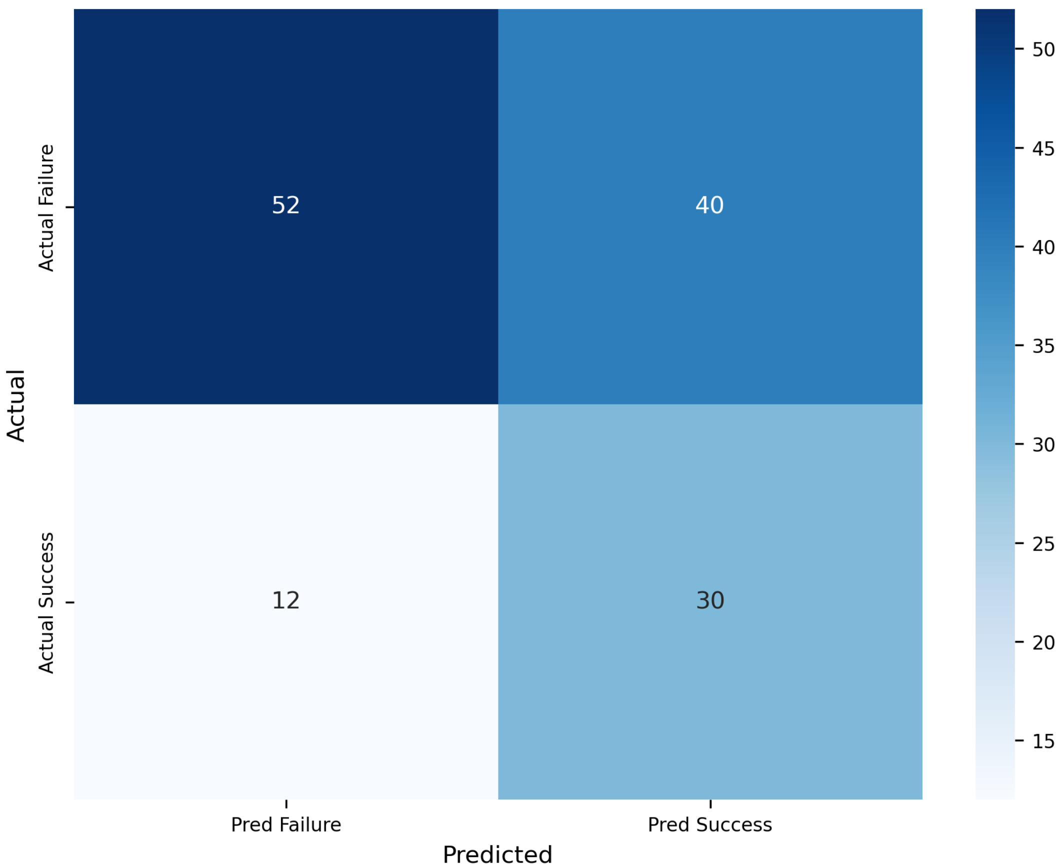

Figure A21: Confusion matrix for Heterogenous GNN on the test dataset.

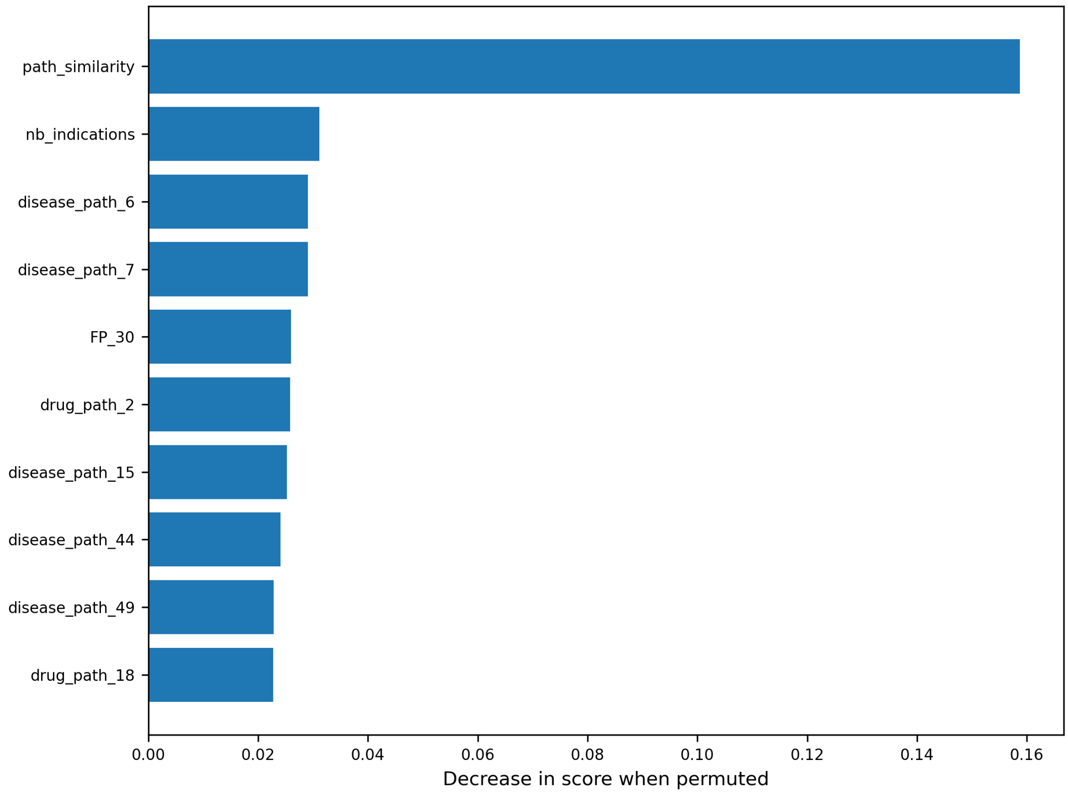

Figure A22: Permutation feature importance for trained XGBoost.

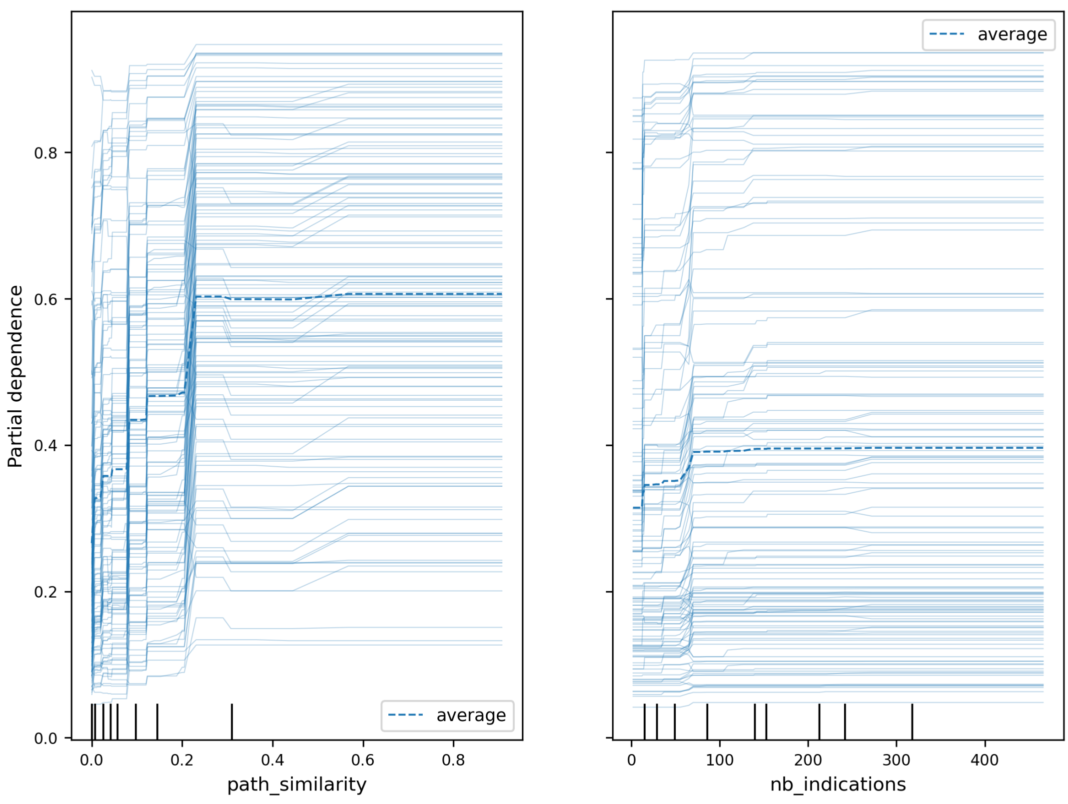

Figure A23: Partial dependence of the two most important features.

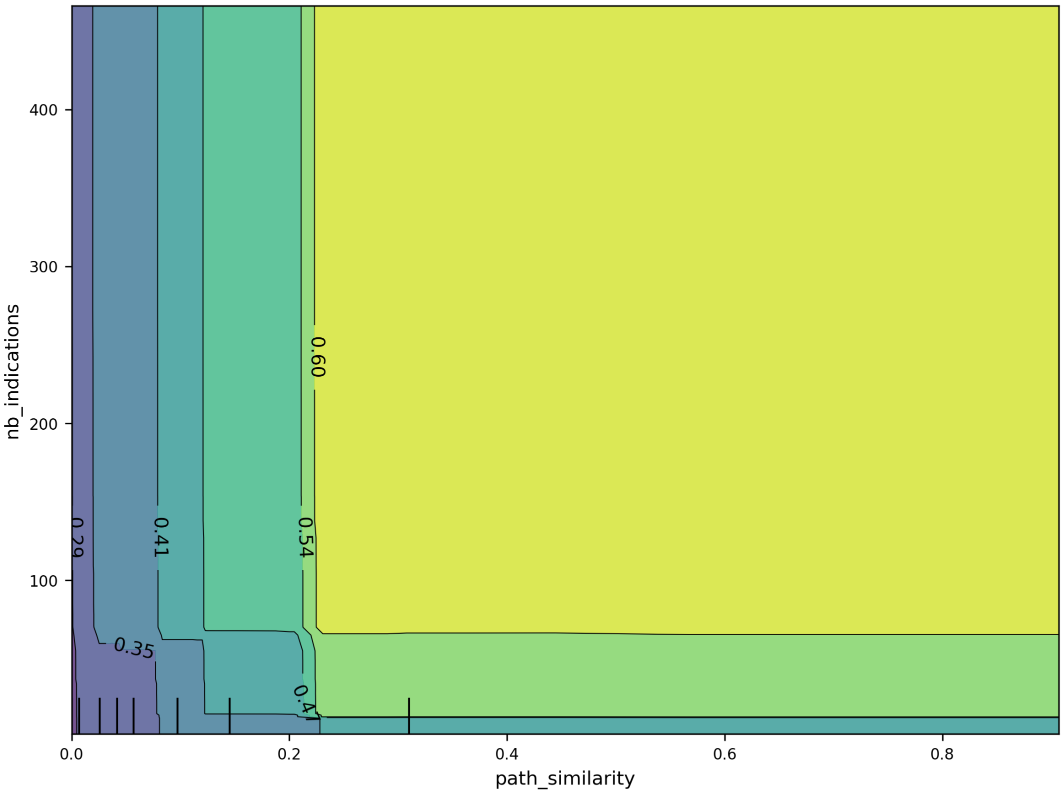

Figure A24: Two-dimensional partial dependence of the two most important features.

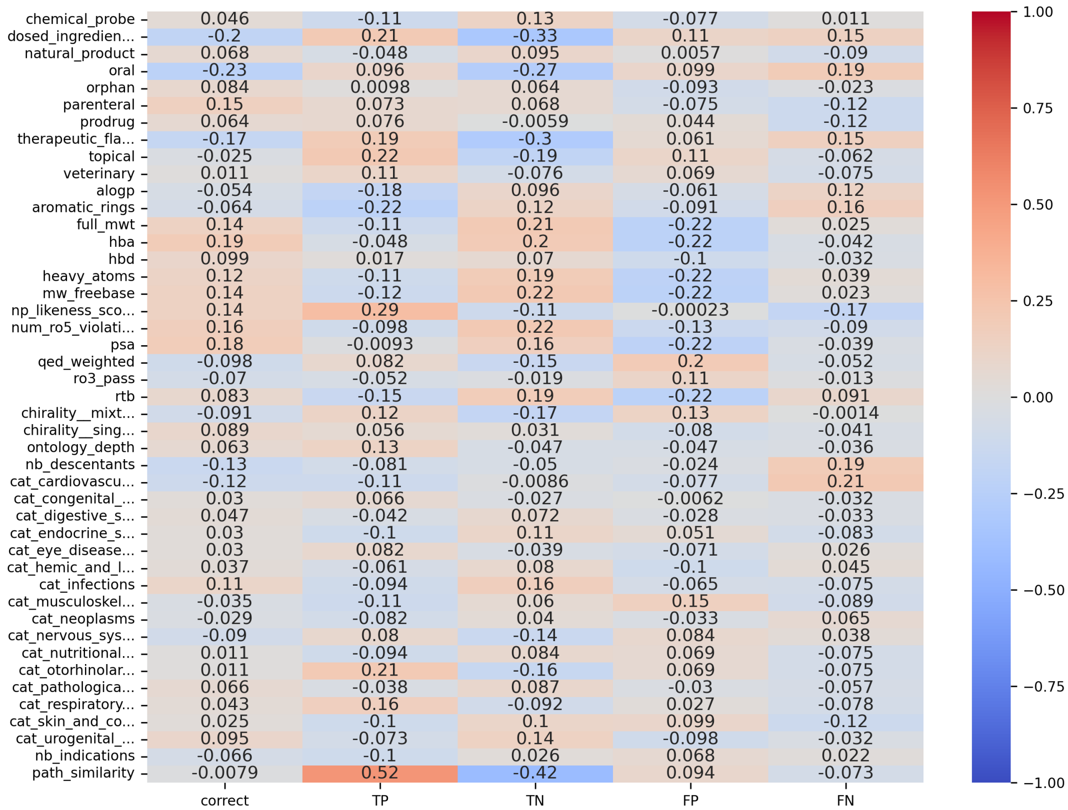

Figure A25: Correlation heatmap of XGBoost test classification results with predictors.

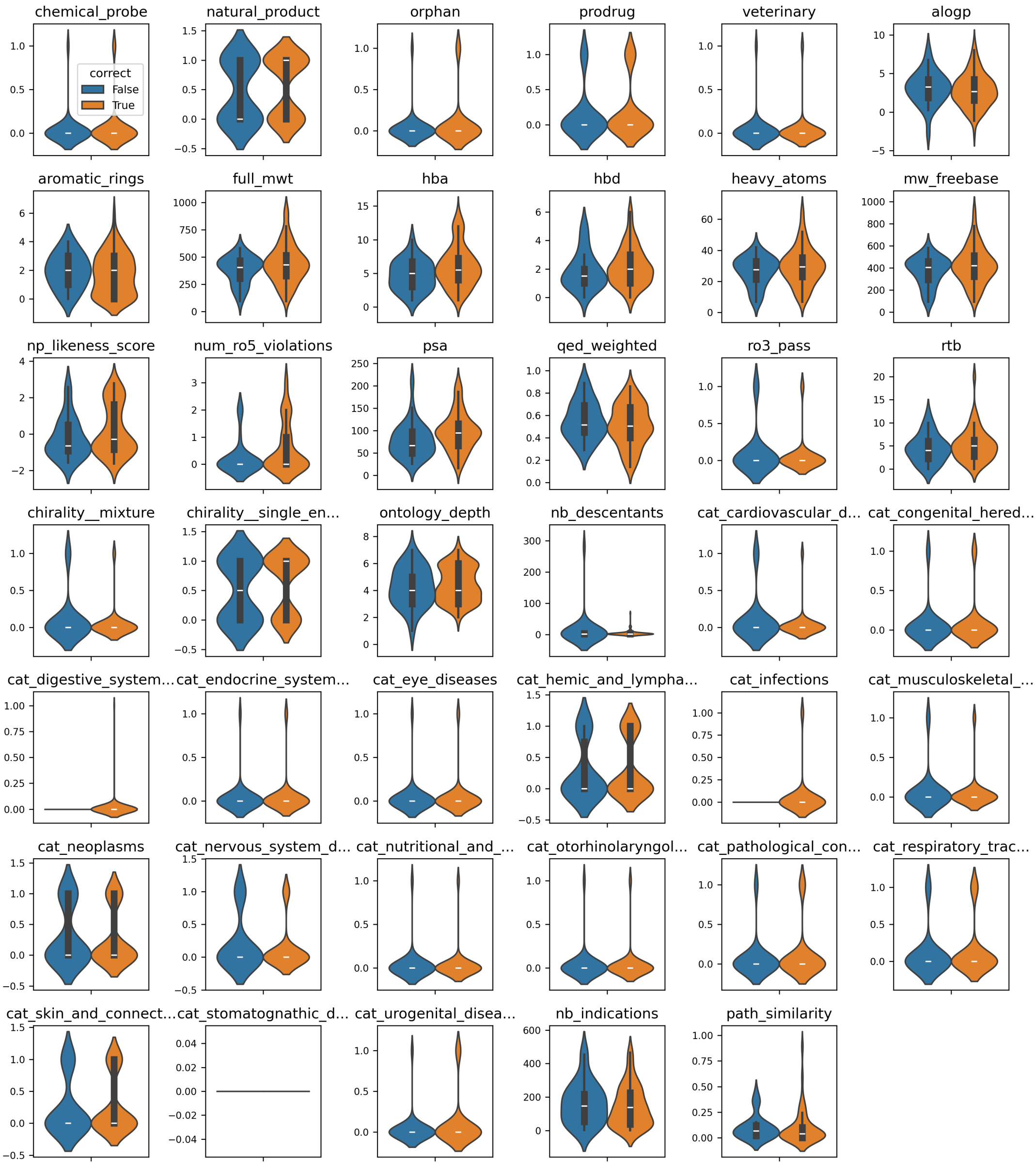

Figure A26: Feature distributions for samples correctly vs. incorrectly classified by XG-Boost.

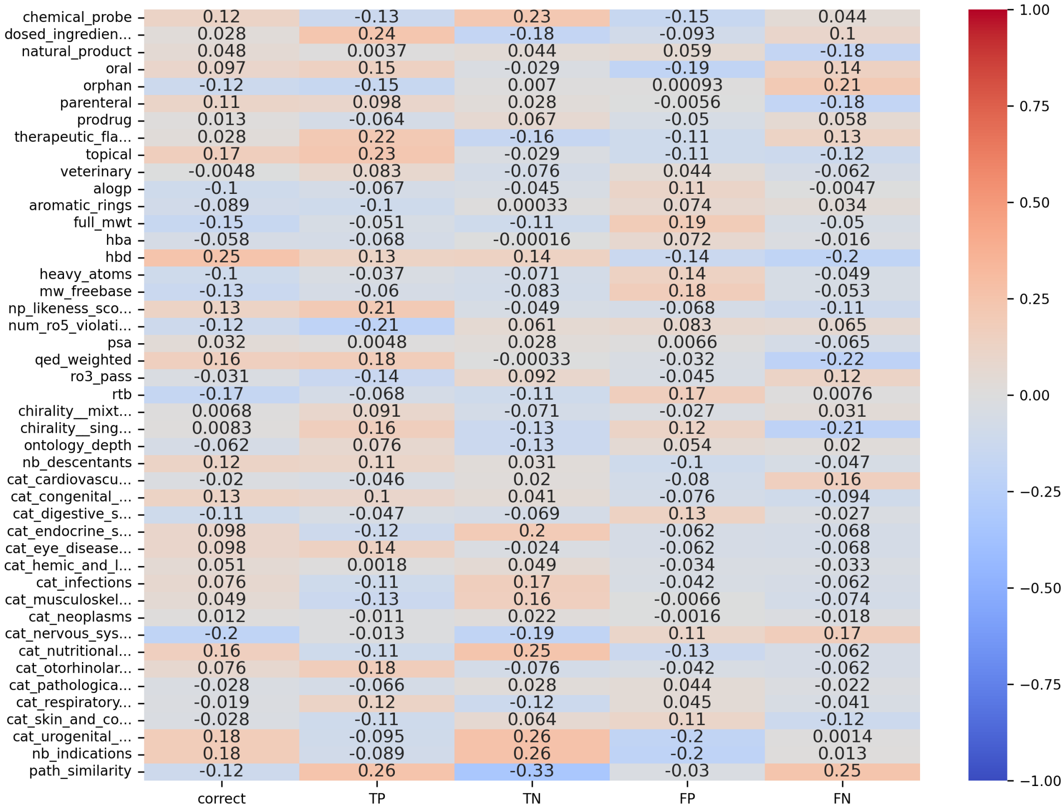

Figure A27: Correlation heatmap of Heterogenous GNN test classification results with predictors.

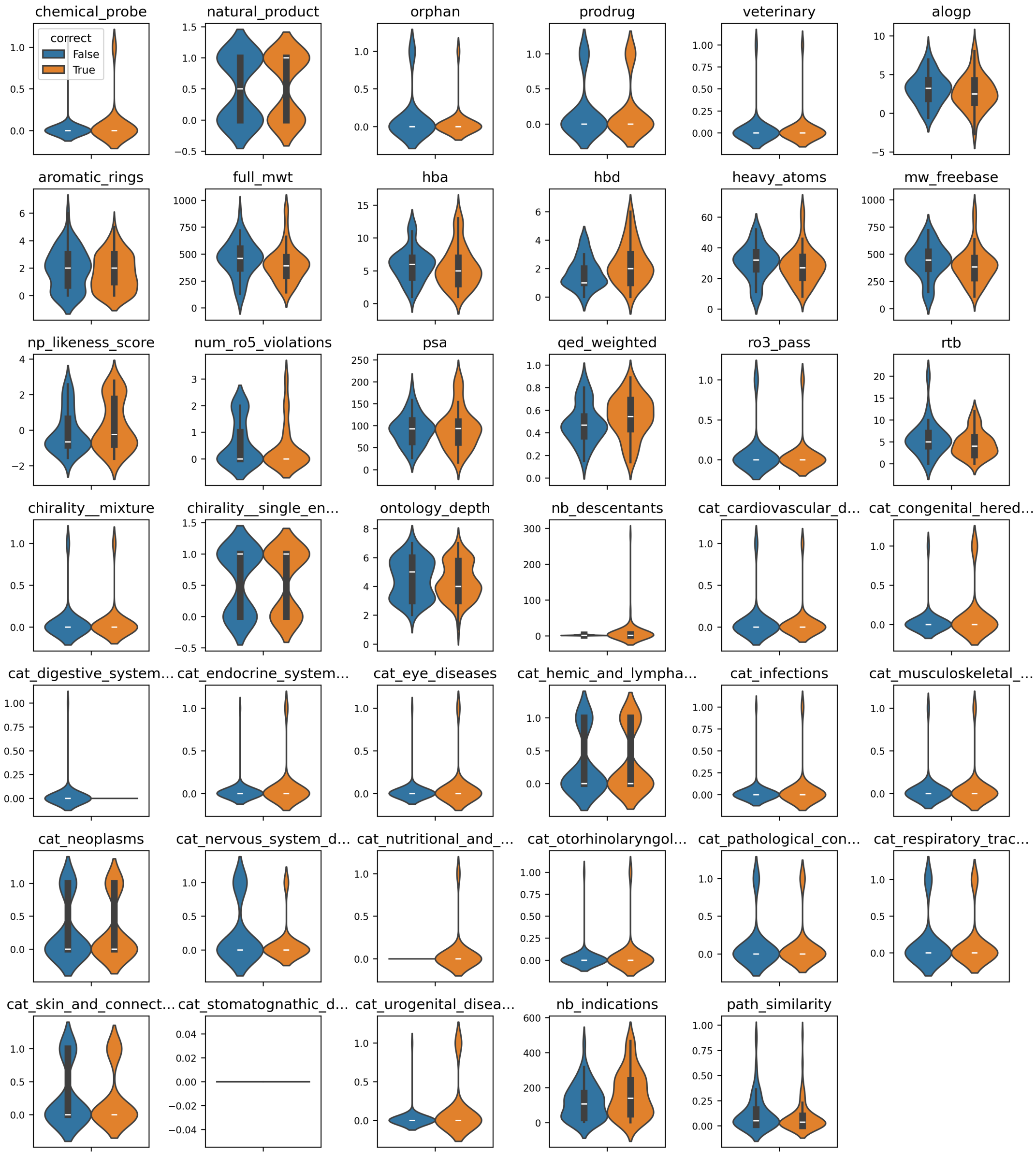

Figure A28: Feature distributions for samples correctly vs. incorrectly classified by Heterogenous GNN.